In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler
from sklearn.cluster import KMeans

# Load World Bank Excel file
file_path = "P_Data_Extract_From_World_Development_Indicators.xlsx"

df = pd.read_excel(file_path, sheet_name="Data")

# Drop empty rows
df = df.dropna(subset=["Series Name"]).copy()

# Reshape from wide to long format
long = df.melt(
    id_vars=["Country Name", "Country Code", "Series Name", "Series Code"],
    var_name="Year",
    value_name="Value"
)

# Clean year column
long["Year"] = long["Year"].astype(str).str.extract(r"(\d{4})")
long = long.dropna(subset=["Year"])
long["Year"] = long["Year"].astype(int)

# Convert values to numbers
long["Value"] = pd.to_numeric(long["Value"], errors="coerce")

# Pivot so each indicator becomes a column
master = long.pivot_table(
    index=["Country Name", "Year"],
    columns="Series Name",
    values="Value"
).reset_index()

# Clean column names
master.columns.name = None

print("Columns in dataset:")
print(master.columns.tolist())

# Rename indicators to shorter names
rename_map = {
    "Individuals using the Internet (% of population)": "internet_users",
    "Unemployment, youth total (% of total labor force ages 15-24) (modeled
    "School enrollment, tertiary (% gross)": "tertiary_enrollment",
    "Employment in services (% of total employment) (modeled ILO estimate)":
    "GDP per capita (current US$)": "gdp_per_capita",
    "Mobile cellular subscriptions (per 100 people)": "mobile_subscriptions"
    "High-technology exports (% of manufactured exports)": "hightech_exports
}

master = master.rename(columns=rename_map)
```

```
# Indicators used for digital readiness
indicators = [
    "internet_users",
    "tertiary_enrollment",
    "gdp_per_capita",
    "mobile_subscriptions",
    "services_employment",
    "hightech_exports"
]

# Keep only indicators that actually exist
available_indicators = [col for col in indicators if col in master.columns]

print("\nAvailable indicators used:")
print(available_indicators)

# Create normalized versions for index
index_df = master.copy()

scaler = MinMaxScaler()

index_df[available_indicators] = scaler.fit_transform(
    index_df[available_indicators].fillna(index_df[available_indicators].mec
)

# Digital Readiness Index
index_df["digital_readiness_index"] = (
    0.25 * index_df.get("internet_users", 0)
    + 0.20 * index_df.get("tertiary_enrollment", 0)
    + 0.20 * index_df.get("gdp_per_capita", 0)
    + 0.15 * index_df.get("mobile_subscriptions", 0)
    + 0.10 * index_df.get("services_employment", 0)
    + 0.10 * index_df.get("hightech_exports", 0)
)

# Save cleaned datasets
master.to_csv("central_asia_digital_indicators_clean.csv", index=False)
index_df.to_csv("central_asia_digital_readiness_index.csv", index=False)

# Latest year ranking
latest = (
    index_df
    .sort_values("Year")
    .groupby("Country Name")
    .tail(1)
    .sort_values("digital_readiness_index", ascending=False)
)

latest_output = latest[[
    "Country Name",
    "Year",
    "digital_readiness_index"
]]

latest_output.to_csv("latest_digital_readiness_ranking.csv", index=False)
```

```
print("\nLatest Digital Readiness Ranking:")
print(latest_output.to_string(index=False))

# Plot digital readiness over time
plt.figure(figsize=(10, 6))

for country in index_df["Country Name"].dropna().unique():
    temp = index_df[index_df["Country Name"] == country]
    plt.plot(
        temp["Year"],
        temp["digital_readiness_index"],
        marker="o",
        label=country
    )

plt.title("Digital Readiness Index Over Time")
plt.xlabel("Year")
plt.ylabel("Digital Readiness Index")
plt.legend()
plt.tight_layout()
plt.savefig("digital_readiness_over_time.png")
plt.show()

# Latest country comparison chart
plt.figure(figsize=(9, 5))

plt.bar(
    latest["Country Name"],
    latest["digital_readiness_index"]
)

plt.title("Latest Digital Readiness Ranking")
plt.xlabel("Country")
plt.ylabel("Digital Readiness Index")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.savefig("latest_digital_readiness_ranking.png")
plt.show()

# Cluster analysis using latest data
cluster_data = latest[["Country Name"] + available_indicators].copy()

X = cluster_data[available_indicators].fillna(
    cluster_data[available_indicators].median()
)

kmeans = KMeans(
    n_clusters=3,
    random_state=42,
    n_init=10
)

cluster_data["cluster"] = kmeans.fit_predict(X)

cluster_data.to_csv("country_digital_clusters.csv", index=False)
```

```
print("\nCountry Clusters:")
print(cluster_data[["Country Name", "cluster"]].to_string(index=False))

# Simple strategic interpretation
tajikistan = latest[latest["Country Name"].str.contains("Tajikistan", case=F

if not tajikistan.empty:
    tajik_score = tajikistan["digital_readiness_index"].iloc[0]
    avg_score = latest["digital_readiness_index"].mean()

    print("\nTajikistan Interpretation:")
    print(f"Tajikistan score: {tajik_score:.3f}")
    print(f"Regional average: {avg_score:.3f}")

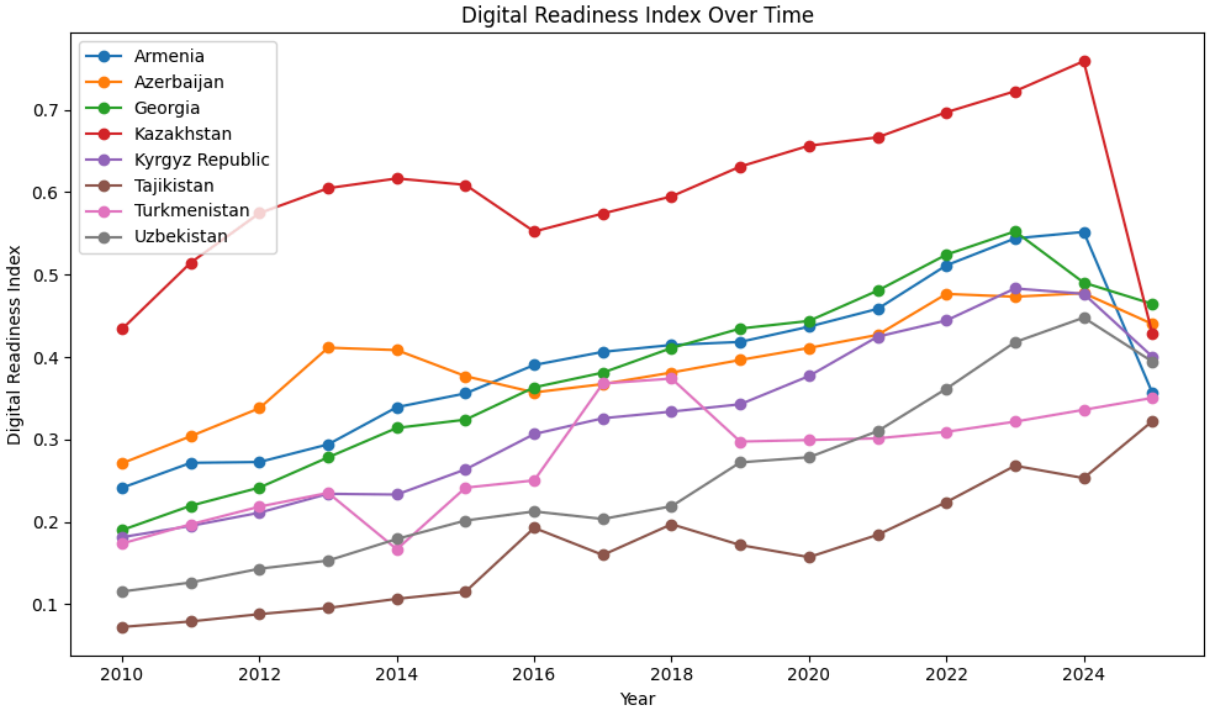
    if tajik_score >= avg_score:
        print("Tajikistan is at or above the peer-country average.")
    else:
        print("Tajikistan is below the peer-country average, suggesting room

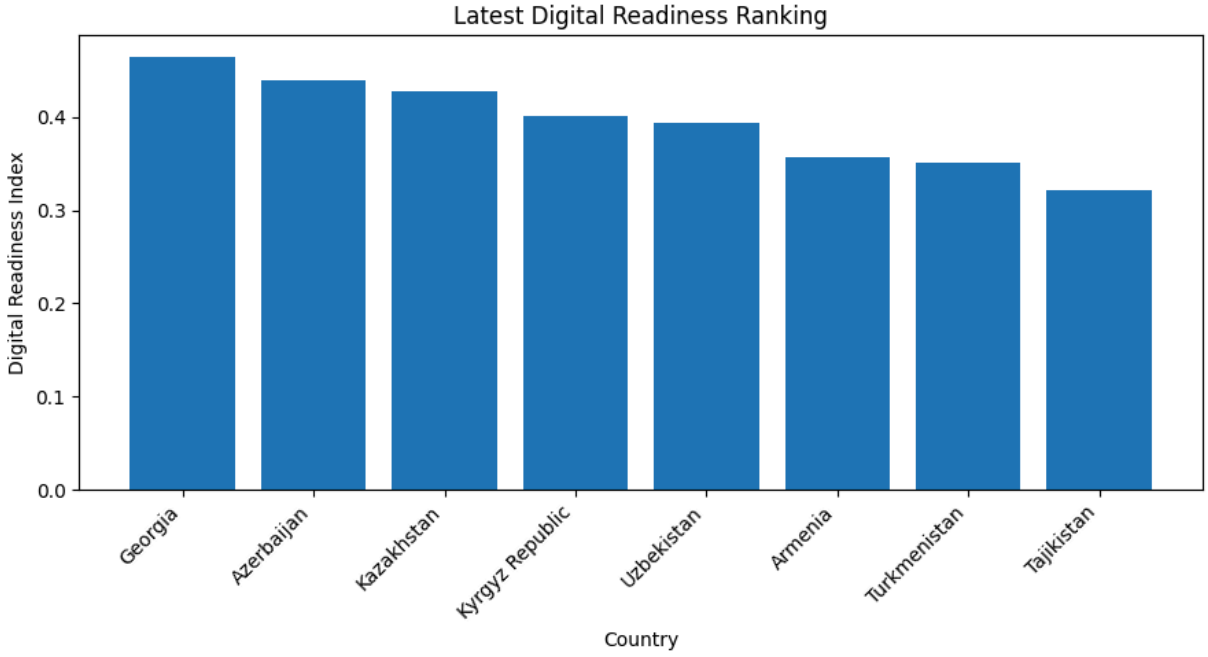
print("\nFiles saved:")
print("central_asia_digital_indicators_clean.csv")
print("central_asia_digital_readiness_index.csv")
print("latest_digital_readiness_ranking.csv")
print("country_digital_clusters.csv")
print("digital_readiness_over_time.png")
print("latest_digital_readiness_ranking.png")
```

```
Columns in dataset:
['Country Name', 'Year', 'Employment in services (% of total employment) (mo
deled ILO estimate)', 'GDP per capita (current US$)', 'High-technology expor
ts (% of manufactured exports)', 'Individuals using the Internet (% of popul
ation)', 'Mobile cellular subscriptions', 'School enrollment, tertiary (% gr
oss)', 'Unemployment, youth total (% of total labor force ages 15-24) (model
ed ILO estimate)']

Available indicators used:
['internet_users', 'tertiary_enrollment', 'gdp_per_capita', 'services_employ
ment', 'hightech_exports']

Latest Digital Readiness Ranking:
   Country Name  Year  digital_readiness_index
        Georgia  2025                 0.464419
     Azerbaijan  2025                 0.440074
     Kazakhstan  2025                 0.427925
Kyrgyz Republic  2025                 0.400428
     Uzbekistan  2025                 0.394394
        Armenia  2025                 0.356442
   Turkmenistan  2025                 0.350404
     Tajikistan  2025                 0.322166
```


Digital Readiness Index Over Time
Armenia
Azerbaijan
Georgia
Kazakhstan
Kyrgyz Republic
Tajikistan
Turkmenistan
Uzbekistan
Digital Readiness Index
0.1
0.2
0.3
0.4
0.5
0.6
0.7
2010
2012
2014
2016
2018
2020
2022
2024
Year


Latest Digital Readiness Ranking
Digital Readiness Index
0.0
0.1
0.2
0.3
0.4
Georgia
Azerbaijan
Kazakhstan
Kyrgyz Republic
Uzbekistan
Armenia
Turkmenistan
Tajikistan
Country

```
Country Clusters:
   Country Name  cluster
        Georgia        2
     Azerbaijan        1
     Kazakhstan        0
Kyrgyz Republic        1
     Uzbekistan        1
        Armenia        1
   Turkmenistan        1
     Tajikistan        1

Tajikistan Interpretation:
Tajikistan score: 0.322
Regional average: 0.395
Tajikistan is below the peer-country average, suggesting room for digital sk
ills expansion.

Files saved:
central_asia_digital_indicators_clean.csv
central_asia_digital_readiness_index.csv
latest_digital_readiness_ranking.csv
country_digital_clusters.csv
digital_readiness_over_time.png
latest_digital_readiness_ranking.png
```

In [ ]: